

# FLUID-ICU data dictionary

> **SIMULATED DATA: NOT REAL PATIENTS.** FLUID-ICU was generated by `R/00_generate_fluid_icu.R` for teaching.

**Study:** FLUID-ICU, a multicentre prospective cohort of adults with sepsis admitted to 4 ICUs (ICU A-D), January 2023 to December 2024.

**Files:** `fluid_icu_raw.csv / .xlsx` (as sent by the sites, 623 rows, used in the cleaning demo) and `fluid_icu_clean.csv / .rds` (620 patients, 32 variables, analysis-ready).

**Main exposure:** cumulative fluid balance at day 3 (`cum_fluid_balance_d3_ml_kg`, `positive_balance_d3`). **Primary outcome:** `died_28d` (with `time_to_event_28d`).

**Pre-specified confounders:** `age`, `sofa`, `lactate_mmol_l`, `vasopressor`, `mech_vent`, `ckd`. **Effect modifier:** `heart_failure`.

In the clean file a missing value is an empty cell (CSV) or `NA` (R). *Missing code (raw)* gives how the raw file recorded a missing value. *Allowed range* is the plausibility check used in `R/01_clean.R`; values outside it are set to missing and queried. DERIVED variables are computed in cleaning and do not exist in the raw file.

| # | Variable                    | Label                             | Type                   | Units / codes                 | Allowed range            | Missing code (raw) | Notes                                                                                                                                                       |
|---|-----------------------------|-----------------------------------|------------------------|-------------------------------|--------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <code>study_id</code>       | Study identifier                  | Character (ID)         | FLU-001 ... FLU-620           | Unique                   | None allowed       | Raw column 'Study ID'. Raw file has 3 IDs entered twice (exact duplicate rows); removed in cleaning.                                                        |
| 2 | <code>site</code>           | Recruiting ICU                    | Categorical (4 levels) | ICU A / ICU B / ICU C / ICU D | 4 levels                 | None allowed       | Raw 'Site' typed several ways ( <code>icu a</code> , <code>ICU-A</code> , <code>ICUA</code> ); standardised. ICU C reported lactate in <code>mg/dL</code> . |
| 3 | <code>admission_date</code> | Date of ICU admission (enrolment) | Date                   | YYYY-MM-DD                    | 2023-01-01 to 2024-12-31 | None allowed       | Raw 'Admission Date' mixes <code>YYYY-MM-DD</code> , <code>DD/M</code> , <code>M/YYYY</code> and <code>DD-Mon-YYYY</code> .                                 |

| #  | Variable         | Label                    | Type                   | Units / codes                                                | Allowed range | Missing code (raw) | Notes                                                                                         |
|----|------------------|--------------------------|------------------------|--------------------------------------------------------------|---------------|--------------------|-----------------------------------------------------------------------------------------------|
| 4  | age              | Age                      | Numeric (integer)      | Years                                                        | 18 to 110     | Blank              | Raw 'Age'. One impossible value (210) set to missing; query to site.                          |
| 5  | sex              | Sex                      | Categorical (2 levels) | Female / Male                                                | 2 levels      | None allowed       | Raw 'Sex' has 8-9 spellings (M, m, male, F, f, female, 'Female').                             |
| 6  | weight_kg        | Body weight on admission | Numeric                | kg (1 decimal)                                               | 30 to 250     | Blank              | Raw 'Weight (kg)'. One impossible value (7) set to missing. Used to compute balance in ml/kg. |
| 7  | diabetes         | Diabetes mellitus        | Binary                 | No / Yes                                                     | No, Yes       | None               | Raw codes Yes/No/Y/N /yes/no/1/0 (ICU B sends 1/0).                                           |
| 8  | ckd              | Chronic kidney disease   | Binary                 | No / Yes                                                     | No, Yes       | None               | Pre-specified confounder. Mixed Yes/No coding in raw file.                                    |
| 9  | heart_failure    | Chronic heart failure    | Binary                 | No / Yes                                                     | No, Yes       | None               | Pre-specified effect modifier (Q1). Mixed coding in raw file.                                 |
| 10 | hypertension     | Hypertension             | Binary                 | No / Yes                                                     | No, Yes       | None               | Mixed coding in raw file.                                                                     |
| 11 | infection_source | Source of infection      | Categorical (5 levels) | Respiratory / Urinary / Abdominal / Skin/soft tissue / Other | 5 levels      | None               | Ordered as listed for tables.                                                                 |

| #  | Variable           | Label                                        | Type                    | Units / codes      | Allowed range | Missing code (raw) | Notes                                                                                                                             |
|----|--------------------|----------------------------------------------|-------------------------|--------------------|---------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 12 | sofa               | SOFA score on ICU admission                  | Numeric (integer score) | Points             | 0 to 24       | None               | Severity; pre-specified confounder. Median 7 (IQR 5-9).                                                                           |
| 13 | apache_ii          | APACHE II score (first 24 h)                 | Numeric (integer score) | Points             | 0 to 71       | -99                | Raw 'APACHE II'; 6 values coded -99 = not recorded.                                                                               |
| 14 | lactate_mmol_l     | Blood lactate on admission                   | Numeric                 | mmol/L (1 decimal) | 0.1 to 30     | Blank, 999, -99    | ~15% missing, more often in less severe patients (MAR). ICU C values in raw file are mg/dL (x 9.01): divided by 9.01 in cleaning. |
| 15 | map_mmhg           | Mean arterial pressure on admission          | Numeric (integer)       | mmHg               | 20 to 200     | 999                | Raw 'MAP (mmHg)'; 4 values coded 999.                                                                                             |
| 16 | vasopressor        | Vasopressor on admission                     | Binary                  | No / Yes           | No, Yes       | None               | Pre-specified confounder. Mixed coding in raw file.                                                                               |
| 17 | mech_vent          | Invasive mechanical ventilation on admission | Binary                  | No / Yes           | No, Yes       | None               | Raw 'Mech vent'. Pre-specified confounder.                                                                                        |
| 18 | fluid_in_d1_ml     | Total fluid input, ICU day 1                 | Numeric (integer)       | ml                 | 0 to 20000    | Blank              | All intravenous and enteral fluid incl. drugs and nutrition. Raw 'Fluid in D1 (ml)'.                                              |
| 19 | fluid_output_d1_ml | Total fluid output, ICU day 1                | Numeric (integer)       | ml                 | 0 to 20000    | Blank              | Urine, drains, ultra filtration; insensible losses not included.                                                                  |

| #  | Variable                   | Label                                                  | Type              | Units / codes     | Allowed range | Missing code (raw) | Notes                                                                                                                                                                       |
|----|----------------------------|--------------------------------------------------------|-------------------|-------------------|---------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20 | fluid_in_d2_ml             | Total fluid input, ICU day 2                           | Numeric (integer) | ml                | 0 to 20000    | Blank              |                                                                                                                                                                             |
| 21 | fluid_output_d2_ml         | Total fluid output, ICU day 2                          | Numeric (integer) | ml                | 0 to 20000    | Blank              | One negative value (impossible) in raw file set to missing.                                                                                                                 |
| 22 | fluid_in_d3_ml             | Total fluid input, ICU day 3                           | Numeric (integer) | ml                | 0 to 20000    | Blank              |                                                                                                                                                                             |
| 23 | fluid_output_d3_ml         | Total fluid output, ICU day 3                          | Numeric (integer) | ml                | 0 to 20000    | Blank              |                                                                                                                                                                             |
| 24 | cum_fluid_balance_d3_ml_kg | Cumulative fluid balance, ICU days 1-3                 | Numeric (DERIVED) | ml/kg (1 decimal) | -150 to 250   | Derived            | = (sum of fluid in d1-3 - sum of fluid out d1-3) / weight_kg. Not in the raw file. Missing for 2 patients (weight or output missing). Main exposure (Q2, Q3: per 10 ml/kg). |
| 25 | positive_balance_d3        | Positive cumulative fluid balance at day 3             | Binary (DERIVED)  | No / Yes          | No, Yes       | Derived            | Yes if cum_fluid_balance_d3_ml_kg > 0. Main exposure for Q1 and Q5.                                                                                                         |
| 26 | aki_stage                  | Acute kidney injury, maximum KDIGO stage, ICU days 1-7 | Ordinal (integer) | 0 = none, 1, 2, 3 | 0 to 3        | None               | Raw 'AKI stage (KDIGO)'.                                                                                                                                                    |
| 27 | aki_any                    | Any acute kidney injury (KDIGO stage >= 1)             | Binary (DERIVED)  | No / Yes          | No, Yes       | Derived            | Yes if aki_stage >= 1. Outcome for Q2.                                                                                                                                      |

| #  | Variable          | Label                                         | Type              | Units / codes    | Allowed range | Missing code (raw) | Notes                                                                                                                             |
|----|-------------------|-----------------------------------------------|-------------------|------------------|---------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 28 | rrt               | Renal replacement therapy by day 28           | Binary            | No / Yes         | No, Yes       | None               | Mixed coding in raw file.                                                                                                         |
| 29 | vent_free_days_28 | Ventilator-free days to day 28                | Numeric (integer) | Days             | 0 to 28       | None               | Days alive and free of invasive ventilation to day 28; 0 if died; 28 if never ventilated and alive. Raw 'VFD 28'. Outcome for Q3. |
| 30 | icu_los_days      | ICU length of stay                            | Numeric           | Days (1 decimal) | > 0           | None               | Right-skewed. For patients who died, stay ends at death. Outcome for Q5 (survivors).                                              |
| 31 | died_28d          | Death from any cause by day 28                | Binary            | No / Yes         | No, Yes       | None               | Primary outcome (Q1, Q4). Mixed coding in raw file.                                                                               |
| 32 | time_to_event_28d | Time from ICU admission to death or censoring | Numeric (integer) | Days             | 1 to 28       | None               | Death day if died; 28 (censored) if alive at day 28. Use with died_28d for Kaplan-Meier and Cox. Raw 'Time to event 28d (days)'.  |

Column names in the raw file are spreadsheet-style (e.g. Weight (kg), MAP (mmHg));  
`janitor::clean_names()` plus four renames give the names above (see `R/01_clean.R`, step 1).